

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.1: Central Processing Unit (CPU) Architecture

Concept 4.1.8: Interrupts and Interrupt Service Routines

Total Questions: 8

Mark Schemes begin on Page 6

Question 1 | Source: 9618_s25_qp_12 (Q1.c)

(c) Explain how an interrupt from an input device will be detected and handled in the F-E cycle.

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..... [4]

Question 2 | Source: 9618_w23_qp_13 (Q9.c)

9 A computer system is designed using the basic Von Neumann model.

Registers and buses are components in the Von Neumann model.

(c) Interrupts can be caused by software programs or hardware devices.

State **one** cause of a software interrupt.

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..... [1]

Question 3 | Source: 9618_w23_qp_13 (Q9.d)

9 A computer system is designed using the basic Von Neumann model.

Registers and buses are components in the Von Neumann model.

- (d) The following statements describe the stages that the CPU performs when an interrupt is detected.

There are **three** missing statements.

Write the letter of the missing statements from the table in the correct place to complete the description.

- 1 At the end of each Fetch-Execute (F-E) cycle, the processor checks if an interrupt flag is set.
- 2
- 3 If the interrupt priority is high enough, the processor saves the current contents of the registers.
- 4
- 5 When servicing of the interrupt is complete, the processor restores the registers.
- 6

Letter	Stage
A	The address of the Interrupt Service (ISR) handling routine is loaded into the Program Counter (PC).
B	Lower priority interrupts are re-enabled.
C	The device causing the interrupt transfers data to the CPU.
D	The processor identifies the source of the interrupt and checks the priority of the interrupt.
E	The ISR is incremented.

[3]

Question 4 | Source: 9618_s23_qp_11 (Q6)

6 An interrupt is generated when a key is pressed on a computer keyboard.

Explain how the computer handles this interrupt.

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..... [5]

Question 5 | Source: 9618_w22_qp_11 (Q5.c)

(c) Describe the purpose of an interrupt in a computer system.

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..... [2]

Question 6 | Source: 9618_w22_qp_11 (Q5.d)

(d) Identify **two** causes of a software interrupt.

1

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2

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..... [2]

Question 7 | Source: 9618_w22_qp_13 (Q4.a.ii)

4 (a) A Central Processing Unit (CPU) contains several special purpose registers and other components.

(ii) State when interrupts are detected during the Fetch-Execute (F-E) cycle.

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..... [1]

Question 8 | Source: 9618_w21_qp_12 (Q7.a)

7 A computer has system software.

(a) The Operating System handles interrupts.

Tick (✓) **one** box in each row to identify whether each event is an example of a hardware interrupt or a software interrupt.

Event	Hardware interrupt	Software interrupt
Buffer full		
Printer is out of paper		
User has pressed a key on the keyboard		
Division by zero		
Power failure		
Stack overflow		

[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_12 (Q1.c)

1(c)	1 mark each to max 4 - At the start / end of FE cycle the interrupt register is checked - The priority of any interrupts waiting is checked - if the priority of the interrupt is higher than the current process - the contents of the registers are stored on the stack - The relevant Interrupt Service Routine (ISR) / interrupt handler is called to process the interrupt - When the ISR has finished, a further check is made for higher priority interrupts - if no more interrupts of higher priority, the register contents are restored and the next FE cycle continues	4
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Mark Scheme for Question 2 | Source: 9618_w23_ms_13 (Q9.c)

9(c)	1 mark for an appropriate example. Examples: - Division by zero - Runtime error - Out of memory bounds - Program requesting an external device / input - Buffer overflow	1
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Mark Scheme for Question 3 | Source: 9618_w23_ms_13 (Q9.d)

9(d)	1 mark for each letter in the correct position (2, 4 and 6) D A B	3
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Mark Scheme for Question 4 | Source: 9618_s23_ms_11 (Q6)

6	1 mark each to max 5 • An interrupt flag is raised in the (interrupt) register • At the end of the current FE cycle // at the start of the next FE cycle • The system checks the interrupt register for higher priority interrupts than current process • If true, it stores the current contents of the registers on the stack • The appropriate interrupt service routine (ISR) for the key press is called • The input data from the keyboard is processed • The contents of the registers are restored from the stack • ... and control is passed back to previous process	5
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Mark Scheme for Question 5 | Source: 9618_w22_ms_11 (Q5.c)

5(c)	1 mark for each bullet point: • to send a signal from a device or process • ... seeking the attention of the processor	2
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Mark Scheme for Question 6 | Source: 9618_w22_ms_11 (Q5.d)

5(d)	1 mark for each bullet point (max 2). For example: • division by zero // runtime error in a program • attempt to access an invalid memory location • array index out of bounds • stack overflow	2
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Mark Scheme for Question 7 | Source: 9618_w22_ms_13 (Q4.a.ii)

4(a)(ii)	after completion of the execute stage // before the cycle begins	1
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Mark Scheme for Question 8 | Source: 9618_w21_ms_12 (Q7.a)

7(a)	1 mark per pair of rows (shaded & unshaded)	3
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